SDS 8102: Advanced Mathematical Statistics

Midterm Examination

Fall 2026

Mohamed bin Zayed University of Artificial Intelligence

Name: _____ **Student ID:** _____

Date: *Oct 14, 2026* **Duration:** *1 hr 30 mins.*

Instructions

1. This examination contains **four questions**. Answer all questions.
2. The examination is graded out of **50 points**. No buffer points.
3. The midterm contributes **35%** of the final course grade.
4. Explain your reasoning as clearly and completely as possible. Credit will be awarded for clear mathematical explanations, even when the final answer is incomplete.
5. Clearly state every theorem or result that you use, and explain why its assumptions are satisfied in the setting of the problem. Merely naming a theorem is not sufficient.
6. Show all important intermediate steps. A correct final answer without adequate justification may receive only partial credit.
7. Mathematically correct alternative arguments are welcome and will receive full credit. If a minor computational error occurs after a correct setup, partial credit may still be awarded.
8. If you are unable to complete one part of a question, you may use the stated conclusion of that part in subsequent parts.
9. Unless otherwise stated, all random variables are defined on a common probability space, and all expectations and probabilities are assumed to be well-defined.

Write your name and student ID on every answer booklet. If you require additional space, clearly indicate where the continuation of your solution appears.

Question	1	2	3	4	Total
Available	12	12	10	16	50
Score					

Question 1 Poisson distribution and UMVUE (12 Points)

Let $n \in \mathbb{N}$ and $X_1, X_2, , \ldots, X_n \overset{i.i.d.}{\sim} \text{Poisson}(\lambda)$ for an unknown parameter $\lambda > 0$.

(a) Find a complete sufficient statistic for λ. Prove both sufficiency and completeness. [5 points]

(b) Find the UMVUE of $\exp(-\lambda)$. Justify your answer fully. [4 points]

(c) Show that $\sqrt{\lambda}$ can not be unbiasedly estimated. [3 points]

Question 2 Gaussian Correlation and Sign Data (12 Points)

Let

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim \mathcal{N}_2\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}\right), \qquad -1 < \rho < 1.$$

(a) A generative model $\mathcal{G}_\rho$ generates independent observations from the preceding bivariate Gaussian distribution. Based on m i.i.d samples from $\mathcal{G}_\rho$, construct a consistent estimator of ρ and find its asymptotic distribution. [5 points]

(b) Suppose, due to a corrupted output layer, the generator now returns only the sign values

$$\big(\operatorname{sign}(X_i), \operatorname{sign}(Y_i)\big), \qquad i = 1, \ldots, m,$$

rather than the values of X_i and Y_i. Using *only these sign observations*, construct a consistent estimator of ρ and find an asymptotic 95% confidence interval. [7 points]

Bonus: Which one is asymptotically more efficient? [No points]

Hint: It is a standard result that $\mathbb{P}_\rho(X > 0, Y > 0) = \frac{1}{4} + \frac{\arcsin(\rho)}{2\pi}$. You may use this result directly.

Question 3 Alice's Personal-Record Problem (10 Points)

Alice has joined a gym and records her best weight lifted each week. Under a deliberately simple model, her weekly performances

$$X_1, X_2, \dots$$

are i.i.d. with a common continuous distribution function F. Alice declares week j to be a *personal-record (PR) week* if she has hit a new max-weight, i.e.,

$$X_j > \max\{X_1, \dots, X_{j-1}\}, \quad \text{for } j > 1.$$

For $j \geq 2$, define

$$I_j = \mathbf{1}\{X_j \text{ is a personal record}\}, \qquad R_N = \sum_{j=2}^{N} I_j,$$

so that R_N is the number of personal records achieved during the first N weeks.

(a) Show that $I_j \sim \text{Bernoulli}(\frac{1}{j})$. This shows that hitting new PR gets harder with time. [2 points]

(b) Prove that

$$\frac{R_N}{\log N} \longrightarrow 1 \qquad \textit{almost surely.}$$

[8 points]

Useful fact (hint). You may use the following version of **Kronecker's lemma**: if $(a_n)_{n\geq 1}$ is increasing with $a_n \to \infty$ and $\sum_{n=1}^{\infty} \frac{x_n}{a_n}$ converges, then $a_n^{-1} \sum_{k=1}^{n} x_k \longrightarrow 0$.

Question 4 Uniform Convergence over Half-spaces (16 Points)

Let

$$\mathcal{F} = \Big\{ \mathbf{1}\{a^\top x + b \le 0\} : a \in \mathbb{R}^2,\ b \in \mathbb{R} \Big\}$$

be the class of indicators of half-spaces in $\mathbb{R}^2$.

(a) Prove that $\mathrm{VCdim}(\mathcal{F}) = 3$. [6 points]

Recall: $\mathrm{VCdim}(\mathcal{F}) = k \iff$ Some set of k-points can be shattered by $\mathcal{F}$, and no $(k+1)$-points can be shattered by $\mathcal{F}$. For this problem, it might be useful to consider two cases: (i) one point lies in the the convex hull of the other three points, (ii) all four points are in convex positions.

(b) Let $X_1, X_2, \ldots$ be i.i.d. observations from an arbitrary distribution P on $\mathbb{R}^2$, and define

$$P_n f = \frac{1}{n} \sum_{i=1}^{n} f(X_i), \qquad Pf = \mathbb{E}[f(X_1)].$$

Prove that there exists a finite constant $C > 0$ such that

$$\mathbb{P}\left(\sup_{f \in \mathcal{F}} |P_n f - Pf| \le C \sqrt{\frac{\log n}{n}} \text{ for all sufficiently large } n \right) = 1.$$

Using the above, conclude that $\mathcal{F}$ is a Glivenko–Cantelli class:

$$\sup_{f \in \mathcal{F}} |P_n f - Pf| \longrightarrow 0 \qquad \text{almost surely.}$$

[10 points]

Hint: Note that $\mathcal{F}$ is a bounded class of functions, and hence, has bounded difference property.